

# AFRILANG Stdlib

## std/c/gameminimax

**Français.** Bibliothèque AFRILANG 'std/c/gameminimax' (niveau complexe, catégorie complexe). Elle expose 7 fonction(s) : minimax, bestMove, evaluate, isTerminal, negamax, depthReached, scoreDiff.

'import "std/c/gameminimax"' · 'use gameminimax'

- 'minimax(board list number, depth number, maximizing bool) → number' - 'bestMove(board list number, depth number) → number' - 'evaluate(board list number) → number' - 'isTerminal(board list number) → bool' - 'negamax(board list number, depth number, color number) → number' - 'depthReached(depth number) → number' - 'scoreDiff(board list number) → number'

**English.** AFRILANG library 'std/c/gameminimax' (tier complex, category complex). Exports 7 function(s): minimax, bestMove, evaluate, isTerminal, negamax, depthReached, scoreDiff.

'import "std/c/gameminimax"' · 'use gameminimax'

- 'minimax(board list number, depth number, maximizing bool) → number' - 'bestMove(board list number, depth number) → number' - 'evaluate(board list number) → number' - 'isTerminal(board list number) → bool' - 'negamax(board list number, depth number, color number) → number' - 'depthReached(depth number) → number' - 'scoreDiff(board list number) → number'

|                       |                      |               |
|-----------------------|----------------------|---------------|
| Catégorie / Category  | Modules complex (c/) |               |
| Niveau / Tier         | complex              | June 16, 2026 |
| Fonctions / Functions | 7                    |               |

## Contents

## Français

### 1. Vue d'ensemble

Bibliothèque AFRILANG 'std/c/gameminimax' (niveau complex, catégorie complex). Elle expose 7 fonction(s) : minimax, bestMove, evaluate, isTerminal, negamax, depthReached, scoreDiff.

```
'import "std/c/gameminimax"' · 'use gameminimax'
```

```
- `minimax(board list number, depth number, maximizing bool) → number`
- `bestMove(board list number, depth number) → number`
- `evaluate(board list number) → number`
- `isTerminal(board list number) → bool`
- `negamax(board list number, depth number, color number) → number`
- `depthReached(depth number) → number`
- `scoreDiff(board list number) → number`
```

### 2. Installation & import

Ajoutez le module à votre programme :

```
import "std/c/gameminimax"
use gameminimax
```

Niveau : complex · Catégorie : Modules complex (c/)  
Domaines avancés – graphes, NLP, réseaux complexes.

### 3. Référence API

- minimax(board list number, depth number, maximizing bool) → number•
- bestMove(board list number, depth number) → number•
- evaluate(board list number) → number•
- isTerminal(board list number) → bool•
- negamax(board list number, depth number, color number) → number•
- depthReached(depth number) → number•
- scoreDiff(board list number) → number

### 4. Exemples d'utilisation

```
import "std/c/gameminimax"
use gameminimax
```

```
create result = minimax(/* args */)
say result
```

```
create result = bestMove(/* args */)
say result
```

```
create result = evaluate(/* args */)
say result
```

```
create result = isTerminal(/* args */)
say result
```

```
create result = negamax(/* args */)
say result
```

```
create result = depthReached(/* args */)
say result
```

## 5. Bonnes pratiques

- Préférez ``import "std/c/gameminimax"`` en tête de fichier.
- Lancez ``afriLang check`` avant de livrer.
- Combinez avec les modules stdlib de la même catégorie.
- Voir `/stdlib/` pour le source live et les démos playground.
- Signalez les problèmes sur GitHub si une export manque.

## 6. Annexe — source

module gameminimax

```
function minimax(board list number, depth number, maximizing bool) returns number
end
```

```
function bestMove(board list number, depth number) returns number
end
```

```
function evaluate(board list number) returns number
end
```

```
function isTerminal(board list number) returns bool
end
```

```
function negamax(board list number, depth number, color number) returns number
end
```

```
function depthReached(depth number) returns number
end
```

```
function scoreDiff(board list number) returns number
end
end
```

## English

### 1. Overview

AFRILANG library 'std/c/gameminimax' (tier complex, category complex). Exports 7 function(s): minimax, bestMove, evaluate, isTerminal, negamax, depthReached, scoreDiff.

```
'import "std/c/gameminimax"' · 'use gameminimax'
```

```
- `minimax(board list number, depth number, maximizing bool) → number`
- `bestMove(board list number, depth number) → number`
- `evaluate(board list number) → number`
- `isTerminal(board list number) → bool`
- `negamax(board list number, depth number, color number) → number`
- `depthReached(depth number) → number`
- `scoreDiff(board list number) → number`
```

### 2. Installation & import

Add the module to your program:

```
import "std/c/gameminimax"
use gameminimax
```

Tier: complex · Category: Complex modules (c/)  
Advanced domains – graphs, NLP, complex networks.

### 3. API reference

- minimax(board list number, depth number, maximizing bool) → number•
- bestMove(board list number, depth number) → number•
- evaluate(board list number) → number•
- isTerminal(board list number) → bool•
- negamax(board list number, depth number, color number) → number•
- depthReached(depth number) → number•
- scoreDiff(board list number) → number

### 4. Usage examples

```
import "std/c/gameminimax"
use gameminimax
```

```
create result = minimax(/* args */)
say result
```

```
create result = bestMove(/* args */)
say result
```

```
create result = evaluate(/* args */)
say result
```

```
create result = isTerminal(/* args */)
say result
```

```
create result = negamax(/* args */)
say result
```

```
create result = depthReached(/* args */)
say result
```

## 5. Best practices

- Prefer ``import "std/c/gameminimax"`` at the top of your file.
- Run ``afrilang check`` before shipping.
- Combine with related stdlib modules from the same category.
- See `/stdlib/` for the live source and playground demos.
- Report issues on GitHub if an export is missing or undocumented.

## 6. Appendix — source

module gameminimax

```
function minimax(board list number, depth number, maximizing bool) returns number
end
```

```
function bestMove(board list number, depth number) returns number
end
```

```
function evaluate(board list number) returns number
end
```

```
function isTerminal(board list number) returns bool
end
```

```
function negamax(board list number, depth number, color number) returns number
end
```

```
function depthReached(depth number) returns number
end
```

```
function scoreDiff(board list number) returns number
end
end
```

## Référence rapide / Quick reference

- Import : `import "std/c/gameminimax"`
- Vérification : `afrilang check`
- Documentation : <https://afrilang.dev/stdlib/std/c/gameminimax/>

## Source (excerpt)

```
module gameminimax

function minimax(board list number, depth number, maximizing bool) returns number
end

function bestMove(board list number, depth number) returns number
end

function evaluate(board list number) returns number
end

function isTerminal(board list number) returns bool
end

function negamax(board list number, depth number, color number) returns number
end

function depthReached(depth number) returns number
end

function scoreDiff(board list number) returns number
end
end
```